Sharp all-eigenvalue Shannon–Stam stability for Gaussian covariance pairs

Candidate partial-result packet

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Status. Candidate substantial partial result; likely valid; human review recommended. The Poincaré-exponent question for arbitrary log-concave inputs remains open here.

1 Source question

Eldan and Mikulincer prove a Shannon–Stam stability estimate for centered log-concave vectors whose constant deteriorates cubically in a normalized Poincaré parameter [1]. They ask whether the cubic general bound or the quadratic uniformly log-concave bound is tight. They also suggest that the minimum-eigenvalue/Poincaré factor should be replaced by a quantity using all covariance eigenvalues. The exact source passage (PDF page 4) is reproduced below.

Theorem 3. *Let X and Y be centered log-concave vectors with σ_X^2, σ_Y^2 denoting the minimal eigenvalues of their covariance matrices. Assume that $\mathrm{Cov}(X) + \mathrm{Cov}(Y) = 2\mathrm{I}_d$ and set $\max\left(\frac{\mathrm{C_p}(X)}{\sigma_X^2}, \frac{\mathrm{C_p}(Y)}{\sigma_Y^2}\right) = \mathrm{C_p}$. Then, if G denotes the standard Gaussian, for every $\lambda \in (0,1)$*

$$\delta_{EPI,\lambda}(X,Y) \geq K\lambda(1-\lambda)\left(\frac{\min(\sigma_Y^2,\sigma_X^2)}{\mathrm{C_p}}\right)^3 (\mathrm{D}(X||G) + \mathrm{D}(Y||G)),$$

where $K > 0$ is a numerical constant, which can be made explicit.

Remark 4. For ξ-uniformly log-concave vectors, we have the relation, $\mathrm{C_p}(X) \leq \frac{1}{\xi}$ (this is a consequence of the Brascamp-Lieb inequality [3], for instance). Thus, considering Corollary 2, one might have expected that the term $\mathrm{C_p^3}$ could have been replaced by $\mathrm{C_p^2}$ in Theorem 3. We do not know if either result is tight.

Remark 5. Bounding the Poincaré constant of an isotropic log-concave measure is the object of the long standing Kannan-Lováz-Simonovits (KLS) conjecture (see [12, 13] for more information). The conjecture asserts that there exists a constant $K > 0$, independent of the dimension, such that for any isotropic log-concave vector X, $\mathrm{C_p}(X) \leq K$. The best known bound is due to Lee and Vempala which showed in [14] that if X is a a d-dimensional log-concave vector, $\mathrm{C_p}(X) = O\left(\sqrt{d}\right)$.

Concerning the assumptions of Theorem 3; note that as the EPI is invariant to linear transformation, there is no loss in generality in assuming $\mathrm{Cov}(X) + \mathrm{Cov}(Y) = 2\mathrm{I}_d$. Remark that $\mathrm{C_p}(X)$ is, approximately, proportional to the maximal eigenvalue of $\mathrm{Cov}(X)$. Thus, for ill-conditioned covariance matrices $\frac{\mathrm{C_p}(X)}{\sigma_X^2}$, $\frac{\mathrm{C_p}(Y)}{\sigma_Y^2}$ will not be on the same scale. It seems plausible to conjecture that the dependence on the minimal eigenvalue and Poicnaré constant could be replaced by a quantity which would take into consideration all eigenvalues.

The result below completely answers the covariance part on the subclass of all centered Gaussian pairs. It shows, in particular, that arbitrarily ill-conditioned Gaussians cannot force either Poincaré exponent.

2 Proof idea

Let the average covariance be C. A common whitening sends the two covariances to $I + A$ and $I - A$. Since these matrices are polynomials in one symmetric matrix, every quantity diagonalizes in the eigenbasis of A. The deficit and the total relative entropy become sums of elementary scalar logarithmic expressions.

For $\lambda \le 1/2$, put $c = 1 - 2\lambda$. The lower estimate is exactly the concavity inequality

$$\log(1 - ca) \ge c \log(1 - a), \qquad 1 - ca = c(1 - a) + (1 - c) \cdot 1.$$

The upper estimate follows because $\log(1 - ca) + c\log(1 + a)$ has its maximum at $a = 0$. Swapping the inputs proves the other half of the parameter range. No spectral estimate or Poincaré inequality is used.

3 Exact formula and sharp stability

For a positive definite matrix C, write G_C for the centered Gaussian with covariance C. The Shannon–Stam deficit is

$$\delta_\lambda(X, Y) = \lambda \mathrm{D}(X \| G_C) + (1 - \lambda)\mathrm{D}(Y \| G_C) - \mathrm{D}\big(\sqrt{\lambda} X + \sqrt{1 - \lambda} Y \,\|G_C\big).$$

The right side is independent of the chosen centered Gaussian reference, because its constant and quadratic terms cancel; it is therefore the same deficit as in the source.

Theorem 1 (sharp Gaussian spectral stability)**.** *Let X and Y be independent centered Gaussian vectors in $\mathbb{R}^d$ with positive definite covariance matrices Σ_X and Σ_Y. Set*

$$C = \frac{\Sigma_X + \Sigma_Y}{2}, \qquad A = \frac{1}{2} C^{-1/2}(\Sigma_X - \Sigma_Y) C^{-1/2},$$

and let $a_1, \dots, a_d \in (-1, 1)$ be the eigenvalues of A. For every $0 < \lambda < 1$,

$$\delta_\lambda(X, Y) = \frac{1}{2} \sum_{i=1}^{d} \big[\log(1 + (2\lambda - 1)a_i) - \lambda \log(1 + a_i) - (1 - \lambda)\log(1 - a_i)\big], \tag{1}$$

$$\mathrm{D}(X \| G_C) + \mathrm{D}(Y \| G_C) = \frac{1}{2} \sum_{i=1}^{d} -\log(1 - a_i^2). \tag{2}$$

Moreover,

$$\min\{\lambda, 1 - \lambda\}[\mathrm{D}(X \| G_C) + \mathrm{D}(Y \| G_C)] \le \delta_\lambda(X, Y) \tag{3}$$

$$\le \max\{\lambda, 1 - \lambda\}[\mathrm{D}(X \| G_C) + \mathrm{D}(Y \| G_C)]. \tag{4}$$

Both constants are optimal, already in dimension one. At $\lambda = 1/2$ there is the identity

$$\delta_{1/2}(X, Y) = \frac{1}{2}[\mathrm{D}(X \| G_C) + \mathrm{D}(Y \| G_C)]$$

for every such covariance pair.

Proof. Apply the common invertible map $C^{-1/2}$. This leaves the entropy deficit unchanged and sends the covariances to $I + A$ and $I - A$. The covariance of $\sqrt{\lambda}X + \sqrt{1-\lambda}Y$ becomes $I + (2\lambda - 1)A$.

For centered Gaussians,

$$\mathrm{D}(G_\Sigma \| G_I) = \frac{1}{2}\{\operatorname{tr}\Sigma - d - \log\det\Sigma\}.$$

The trace terms cancel in the deficit. Diagonalizing A gives (1). Adding the two input entropies also cancels the trace terms and gives

$$-\frac{1}{2}\log\det((I+A)(I-A)) = \frac{1}{2}\sum_i -\log(1-a_i^2),$$

which is (2).

It remains to prove a scalar statement. Define

$$\begin{aligned} f_\lambda(a) &= \log(1+(2\lambda-1)a) - \lambda\log(1+a) - (1-\lambda)\log(1-a), \\ d(a) &= -\log(1-a^2). \end{aligned}$$

Suppose first that $0 < \lambda \le 1/2$ and put $c = 1 - 2\lambda \in [0, 1)$. Then

$$f_\lambda(a) - \lambda d(a) = \log(1-ca) - c\log(1-a) \ge 0$$

by concavity of the logarithm and $1 - ca = c(1-a) + (1-c)\cdot 1$. On the other hand,

$$f_\lambda(a) - (1-\lambda)d(a) = h_c(a) := \log(1-ca) + c\log(1+a).$$

Here

$$h_c'(a) = \frac{-c(1+c)a}{(1-ca)(1+a)},$$

so h_c attains its maximum $h_c(0) = 0$ at zero. Hence $\lambda d(a) \le f_\lambda(a) \le (1-\lambda)d(a)$. For $\lambda \ge 1/2$, use $f_\lambda(a) = f_{1-\lambda}(-a)$ and $d(-a) = d(a)$. Summing the scalar inequalities proves (3)–(4).

If $\lambda < 1/2$, the ratio $f_\lambda(a)/d(a)$ tends to λ as $a \downarrow -1$ and to $1-\lambda$ as $a \uparrow 1$. The limits are reversed when $\lambda > 1/2$. Thus both constants are sharp in dimension one. When $\lambda = 1/2$, the scalar lower and upper bounds coincide, proving the stated identity. □

Corollary 1 (source normalization)**.** *If $\Sigma_X + \Sigma_Y = 2I$, then $C = I$ and Theorem 1 uses exactly the source's standard Gaussian G. In particular,*

$$\delta_{EPI,\lambda}(X,Y) \ge \min\{\lambda, 1-\lambda\}\,[\mathrm{D}(X\|G) + \mathrm{D}(Y\|G)],$$

uniformly in the dimension, the smallest covariance eigenvalue, and the Gaussian Poincaré constants.

4 Scope and obstruction to a full upgrade

The theorem treats every positive definite Gaussian covariance pair, not only commuting covariances assumed in advance: after average whitening they automatically have the form $I \pm A$. Equations (1) and (2) use every eigenvalue and remain sharp as one eigenvalue tends to degeneracy.

For non-Gaussian inputs, entropy bookkeeping separates the deficit into the Gaussian covariance deficit plus

$$\lambda\mathrm{D}(X\|G_X) + (1-\lambda)\mathrm{D}(Y\|G_Y) - \mathrm{D}(Z\|G_Z),$$

where each $G_\bullet$ matches the covariance of its variable and $Z = \sqrt{\lambda}X + \sqrt{1-\lambda}Y$. This residual has no suitable sign from ordinary convolution data processing when the input covariances differ. Eight focused routes—including perturbation, heat flow, same-law polarization, anisotropic preconditioning, and product extremizers—all return to control of this nonlinear two-law interaction. They are documented in the associated attempt note.

A bounded novelty search on 11 August 2026 used the exact arXiv identifier and question, Gaussian Shannon–Stam covariance stability, log-determinant formulas, Poincaré-exponent phrases, and later equality/stability work. It found the source and adjacent qualitative or regularized results, but no exact match for Theorem 1 and no later resolution of the general exponent question. Because the Gaussian argument is elementary and may be implicit in matrix-entropy literature, novelty confidence is moderate.

Human-review recommendation. Audit the reference-Gaussian cancellation and the two scalar logarithmic inequalities. These are the only substantive steps. Also check whether the sharp two-sided Gaussian formula is already recorded under a different log-determinant Jensen-gap terminology.

References

[1] R. Eldan and D. Mikulincer, *Stability of the Shannon–Stam inequality via the Föllmer process*, ALEA Lat. Am. J. Probab. Math. Stat. **17** (2020), 487–507; arXiv:1903.07140.